

COWRA, Australia

WMO: 957210

Lat: 33.8450S

Lon: 148.6497E

Elev: 974

StdP: 14.19

Time Zone: 10.00 (AUS)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.1	32.1	23.9	18.7	62.2	27.0	21.7	54.0	20.6	53.9	18.5	52.8	1.7	150	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.9	100.1	67.2	96.3	66.6	92.8	65.5	71.8	87.0	70.4	85.3	69.2	83.8	11.9	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.4	108.3	74.5	66.6	101.3	74.4	64.7	94.9	74.0	36.2	86.4	34.9	85.8	33.9	82.8	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.9	18.2	16.0	DB	26.5	105.8	2.0	3.4	25.1	108.3	24.0	110.3	22.9	112.2	21.4	114.6
			WB	25.9	74.2	2.0	1.9	24.4	75.6	23.2	76.7	22.1	77.8	20.6	79.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.1	77.7	75.1	69.7	61.8	53.4	48.8	46.9	48.2	53.3	60.4	66.9	72.2	(6)
	DBStd	11.90	6.27	5.75	5.85	5.55	4.80	4.38	3.91	3.96	4.74	6.44	6.54	6.19	(7)
	HDD50	312	0	0	0	2	21	74	113	83	17	2	0	0	(8)
	HDD65	2669	1	2	24	122	359	487	563	522	352	174	51	12	(9)
	CDD50	4370	859	701	612	356	127	37	16	27	116	323	508	688	(10)
	CDD65	1249	395	283	171	26	0	0	0	0	1	30	109	234	(11)
	CDH74	15809	5051	3222	1780	402	7	0	0	0	69	627	1608	3043	(12)
	CDH80	7517	2802	1593	694	93	0	0	0	0	7	183	673	1472	(13)
(14) Wind	WSAvg	6.4	7.6	7.2	6.6	5.5	5.0	5.3	5.5	5.9	6.6	6.6	7.4	7.5	(14)
(15) Precipitation	PrecAvg	23.4	2.5	1.9	1.6	1.8	1.9	1.6	2.1	2.0	2.1	2.3	2.0	1.9	(15)
	PrecMax	38.0	9.1	6.7	6.4	10.9	6.5	3.8	6.1	4.0	5.2	4.9	4.5	7.7	(16)
	PrecMin	8.5	0.0	0.1	0.0	0.0	0.0	0.1	0.3	0.2	0.4	0.0	0.2	0.0	(17)
	PrecStd	7.1	2.3	1.8	1.4	2.1	1.5	0.9	1.3	1.0	1.2	1.4	1.2	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	105.4	103.4	95.0	88.4	74.9	66.7	65.1	70.1	80.4	89.8	98.0	102.2	(19)
		MCWB	68.5	69.2	66.2	63.1	58.8	54.3	54.9	56.6	56.5	61.5	64.4	65.3	(20)
	2%	DB	100.8	98.2	90.7	82.2	71.2	62.7	61.3	65.3	75.5	85.1	92.0	96.2	(21)
		MCWB	67.5	67.4	64.6	61.2	57.4	53.4	52.5	53.5	57.0	60.9	63.0	66.0	(22)
	5%	DB	97.3	93.3	87.2	78.3	68.2	60.3	58.8	62.4	71.1	81.0	87.8	92.4	(23)
		MCWB	67.5	65.9	64.4	60.1	56.3	52.7	51.3	52.2	55.5	59.5	62.4	64.5	(24)
	10%	DB	93.5	89.4	83.8	75.2	65.0	57.9	56.2	59.4	67.2	76.9	83.1	88.4	(25)
		MCWB	66.9	65.7	63.9	58.8	54.7	52.0	49.7	50.7	54.4	58.8	61.2	63.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.8	74.0	71.0	66.4	62.5	57.2	59.0	58.2	62.1	66.5	69.8	71.9	(27)
		MCDB	89.4	86.2	81.9	76.4	70.0	60.1	61.1	65.6	72.5	76.9	82.7	83.8	(28)
	2%	WB	72.0	71.8	69.1	64.2	59.4	55.4	54.6	55.4	59.6	64.4	67.7	69.8	(29)
		MCDB	89.0	86.6	80.5	74.9	67.6	59.5	58.9	62.6	69.2	76.7	80.8	82.9	(30)
	5%	WB	70.7	70.1	67.4	62.4	57.4	53.9	52.2	53.2	57.6	62.6	65.8	68.0	(31)
		MCDB	87.8	83.2	79.4	74.3	65.9	58.9	56.8	60.1	67.3	75.2	79.1	82.3	(32)
	10%	WB	69.4	68.6	65.8	60.6	55.6	52.5	50.3	51.2	55.5	60.9	64.1	66.4	(33)
		MCDB	86.4	82.3	78.3	72.5	63.8	57.4	55.3	58.6	64.7	73.3	76.9	81.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	26.9	25.1	24.0	25.4	23.3	18.2	19.4	22.2	26.3	28.3	27.0	27.1	(35)
		MCDBR	32.2	31.9	29.5	30.3	27.6	22.0	24.2	28.0	34.9	35.0	33.9	33.3	(36)
	5% WB	MCWBR	9.5	10.0	10.8	13.2	15.1	13.6	15.7	17.0	18.4	15.1	11.7	10.5	(37)
		MCWBR	25.4	22.2	22.5	24.4	24.0	18.5	20.2	24.2	27.5	27.7	24.7	24.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.350	0.346	0.325	0.304	0.291	0.284	0.279	0.287	0.317	0.330	0.345	0.339	(40)
		taud	2.487	2.528	2.555	2.584	2.575	2.609	2.613	2.573	2.473	2.458	2.466	2.506	(41)
	Ebn at Noon		313	308	303	293	280	275	282	295	301	309	313	318	(42)
		Edh at Noon	37	34	31	26	23	21	22	26	33	36	37	36	(43)
(44) All-Sky Solar Radiation	RadAvg		2401	2076	1753	1367	991	750	840	1128	1581	1980	2195	2421	(44)
	RadStd		133	160	133	100	64	66	56	90	130	159	186	117	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)	
(47)	Regional (24 neighbors)	+0.81	N/A	N/A	+1.85	N/A	N/A	N/A	-124	+277	+148	(47)

Nomenclature: See separate page